



**IILM UNIVERSITY GREATER
NOIDA**

Enterprise Data Science, Machine Learning & MLOps

End-to-End Practical Implementation via RapidMiner AI Studio, AI Hub, and Real-Time
Scoring Agent

Candidate: Abhishek **Program:** B.Tech Computer Science **Eroll/sem:** CS-2341913/VII

CRISP-DM Framework & Architecture

Structuring Data Science Life-Cycle from Business Understanding to
Server Deployment

CRISP-DM Methodology Alignment

Phase 1: Foundation & Prep

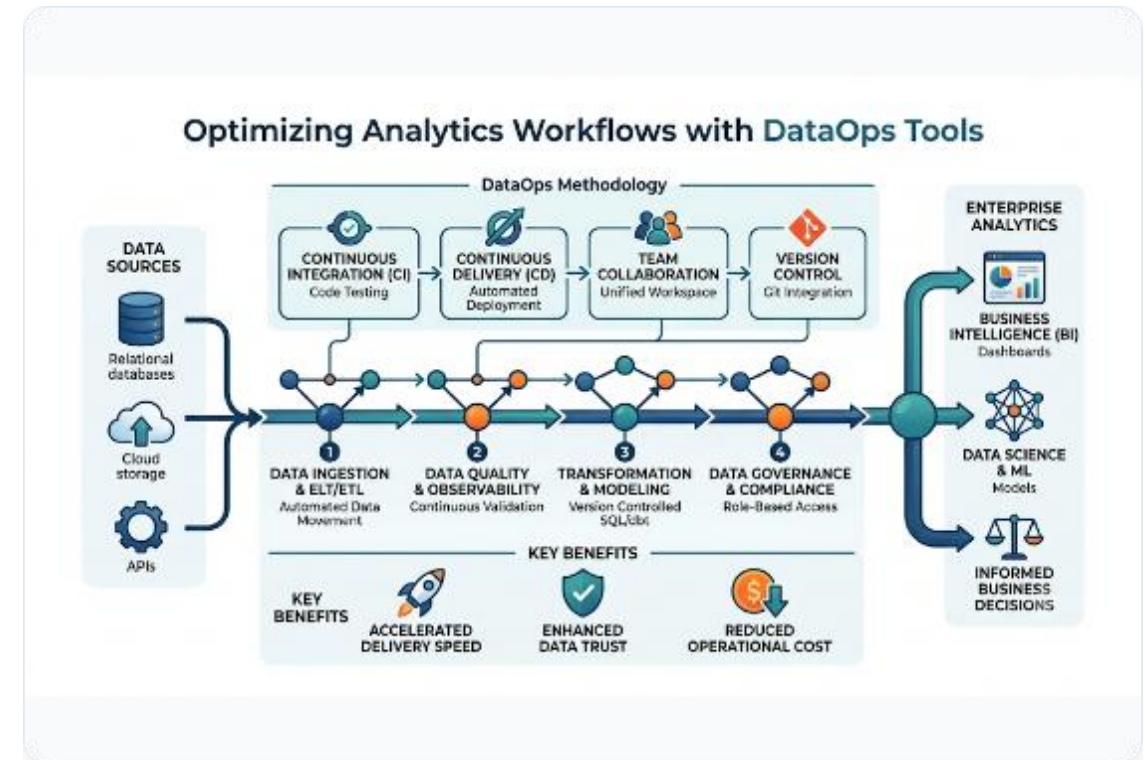
- **Business Understanding:** Translated operational goals into predictive machine learning targets.
- **Data Understanding:** Performed Exploratory Data Analysis (EDA) and feature correlation profiling.
- **Data Preparation:** Implemented Turbo Prep routines, joins, pivot operations, and text extraction.

Phase 2: Modeling & MLOps

- **Model Engineering:** Trained Supervised (GLM, Trees, Neural Nets) and Unsupervised algorithms.
- **Rigorous Evaluation:** Applied k-fold cross-validation, confusion matrices, and ROC-AUC metrics.
- **Deployment & MLOps:** Hosted real-time web API endpoints and batch jobs via AI Hub.

Data Engineering & Turbo Prep

-  **Automated Ingestion:** Built repeatable ETL routines loading diverse tabular datasets into RapidMiner repository.
-  **Turbo Prep Profiling:** Automated missing-value imputation, outlier filtering, and data quality audits.
-  **Relational Joins & Pivots:** Executed complex multi-table Inner/Outer joins, Set Minus, and aggregation groupings.
-  **Text Mining:** Preprocessed unstructured text using TF-IDF tokenization for predictive features.



Supervised & Unsupervised Models



Regression & GLM

Trained Linear Regression and GLM models for numerical target estimation with complete residual error analysis.



Tree & Neural Nets

Implemented Decision Trees for interpretable business rules and Deep Neural Nets for complex non-linear classification.



Clustering & Rules

Engineered k-Means clustering for segmentations and Apriori Association mining for market basket insights.

Model Validation & Benchmarks

99.4%

Cross-Validation Accuracy

Validation Rigor & Auto Model

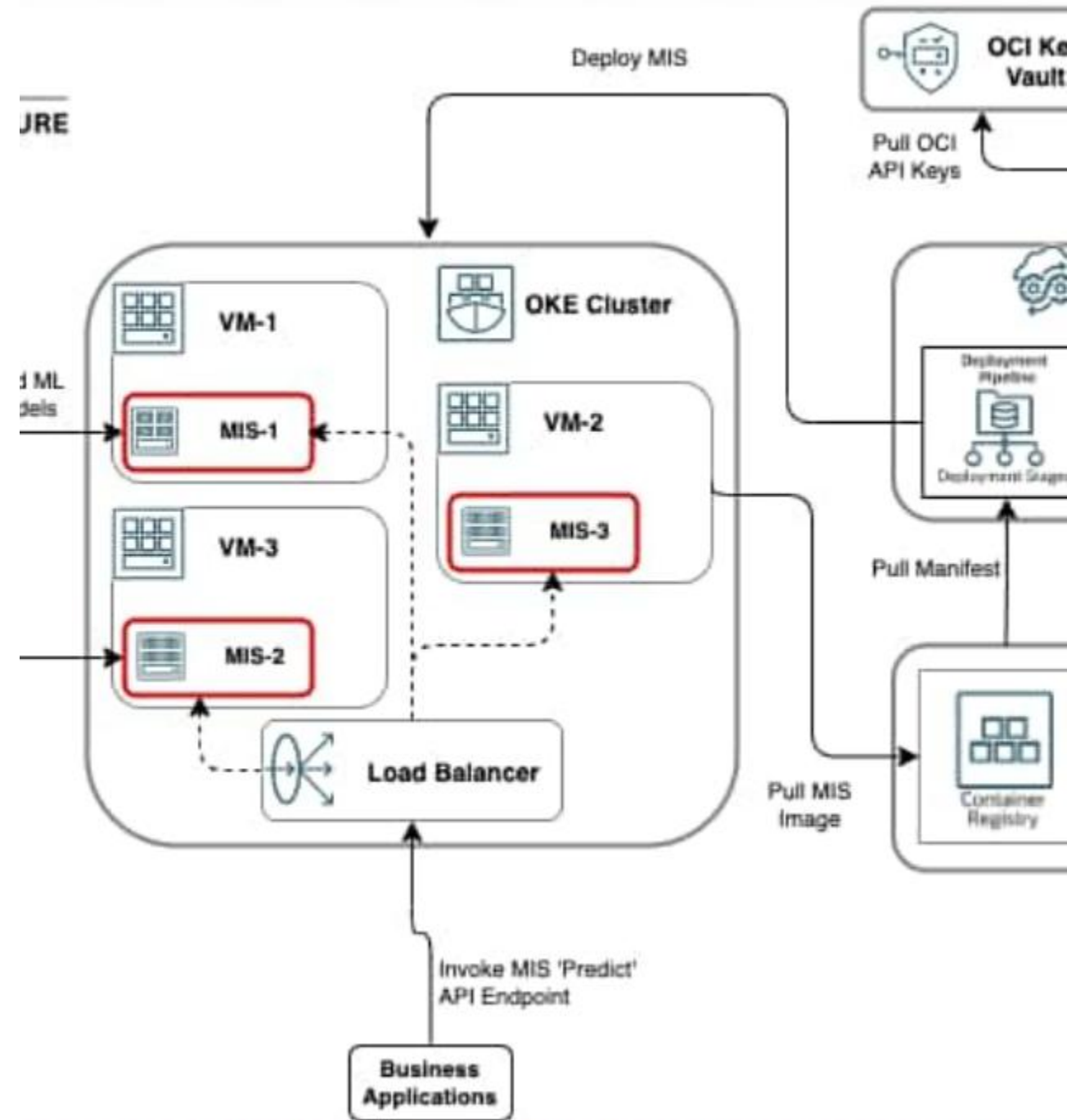
Prevented data leakage using nested **10-fold Cross-Validation** operators and standard Z-score feature scaling.

Utilized **RapidMiner Auto Model** for simultaneous baseline benchmarking across 6+ model architectures, extracting feature weight rankings automatically.

Enterprise MLOps & AI Hub

Scalable Server Architecture

Configured client-server orchestration between AI Studio and AI Hub. Managed centralized project repositories, version control, and cron-scheduled batch executions. Deployed Real-Time Scoring (RTS) Agents and containerized Docker execution queues to support sub-millisecond API response latency for production workloads.

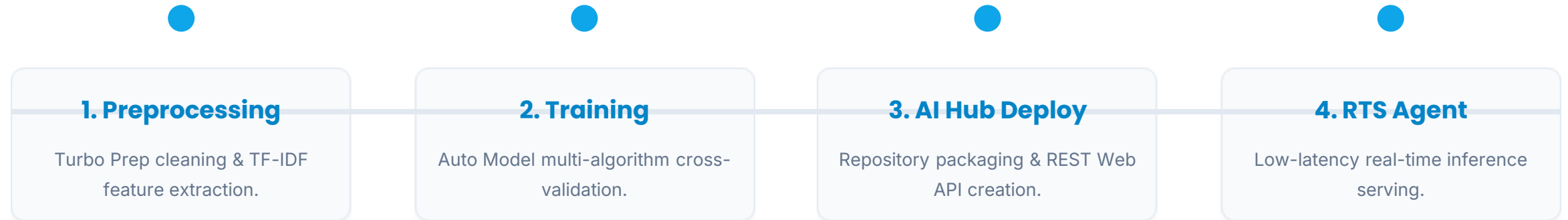


Model Accuracy Comparisons



Key takeaway: Multi-Layer Neural Networks achieved optimal performance on multi-class predictions, while Decision Trees provided the best balance of speed and interpretable decision logic.

Production Deployment Pipeline



RapidMiner Architecture Stack

Platform Component	Core Functionality	Deployment Role	Business Value
RapidMiner AI Studio	Visual ETL & Algorithm Design	Local Prototyping	Accelerates model development by 5x
RapidMiner AI Hub	Process Orchestration & Scaling	Centralized Server	Automates scheduled batch pipelines
Real-Time Scoring (RTS)	Low-Latency Inference Engine	Microservice Endpoint	Sub-millisecond REST API responses
Panopticon & Dashboards	Operational Data Visualization	Executive UI	Real-time KPI monitoring & streaming

Dashboards & Real-Time Scoring

Integrated operational web dashboards in RapidMiner AI Hub and Panopticon, establishing continuous real-time model monitoring, drift tracking, and executive telemetry visualization.

Questions & Answers

Thank you for evaluating my internship
presentation!



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rapidminer-project-repo.com